Chapter 1

Itô's Lemma: Derivations

This note supplies the arguments Chapter 14 states without proof: why $dW^2 = dt$, how Itô's lemma follows, and why a change of measure alters the drift of a diffusion but not its volatility. The level is that of a careful heuristic rather than a measure-theoretic treatment; readers wanting the latter should consult the references at the end.

1.1 The Quadratic Variation of a Wiener Process

Everything rests on one fact: over a fixed interval, the sum of *squared* increments of a Wiener process converges to the length of the interval, and does so with vanishing randomness.

Partition $[0, T]$ into n intervals of length $\Delta t = T/n$, and write ΔW_i for the increment over the ith. From the definition, $\Delta W_i = \varepsilon_i \sqrt{\Delta t}$ with ε_i independent standard normals. Consider

$$Q_n = \sum_{i=1}^{n} \Delta W_i^2 = \Delta t \sum_{i=1}^{n} \varepsilon_i^2 \tag{TN14.1}$$

The mean. Since $\mathbb{E}[\varepsilon_i^2] = 1$,

$$\mathbb{E}[Q_n] = \Delta t \cdot n = T \tag{TN14.2}$$

independently of n.

The variance. For a standard normal, $\mathbb{E}[\varepsilon^4] = 3$, so $\mathrm{Var}(\varepsilon_i^2) = 3 - 1 = 2$. The ε_i are independent, so

$$\mathrm{Var}(Q_n) = \Delta t^2 \cdot n \cdot 2 = 2\,\frac{T^2}{n^2} \cdot n = \frac{2T^2}{n} \tag{TN14.3}$$

As $n \to \infty$ the variance vanishes while the mean stays at T. Therefore

$$\sum_{i=1}^{n} \Delta W_i^2 \;\longrightarrow\; T \quad \text{in mean square} \tag{TN14.4}$$

The limit is not a random variable. This is the content of the informal statement $dW^2 = dt$: squared increments accumulate deterministically, at rate one per unit time.

It is worth contrasting this with the increments themselves. By the same partition, $\sum_i \Delta W_i = W(T) - W(0) \sim N(0, T)$, which remains random however fine the partition. Squaring destroys the randomness; adding does not.

Why ordinary calculus fails. Consider instead a smooth function $x(t)$ with bounded derivative. Then $\Delta x_i = O(\Delta t)$, so $\sum_i \Delta x_i^2 = O(n\,\Delta t^2) = O(\Delta t) \to 0$. For smooth functions the quadratic variation is *zero*, which is precisely why second-order terms are discarded in ordinary Taylor expansions. A Wiener path has $\Delta W_i = O(\sqrt{\Delta t})$, so its quadratic variation is $O(n\,\Delta t) = O(T)$—finite and non-zero. *The whole difference between the two calculi is the difference between $O(\Delta t)$ and $O(\sqrt{\Delta t})$ increments.*

Two corollaries follow by the same argument and are used below:

$$\sum_i \Delta W_i\,\Delta t \to 0, \qquad \sum_i \Delta t^2 \to 0 \tag{TN14.5}$$

so in the shorthand of Chapter 14, $dW\,dt = 0$ and $dt^2 = 0$, while $dW^2 = dt$ alone survives.

1.2 Itô's Lemma

Let S follow the Itô process

$$dS = a(S,t)\,dt + b(S,t)\,dW \tag{TN14.6}$$

and let $f(S,t)$ be twice differentiable in S and once in t. Expand f to second order:

$$\Delta f = \frac{\partial f}{\partial S}\Delta S + \frac{\partial f}{\partial t}\Delta t + \frac{1}{2}\frac{\partial^2 f}{\partial S^2}\Delta S^2 + \frac{\partial^2 f}{\partial S\,\partial t}\Delta S\,\Delta t + \frac{1}{2}\frac{\partial^2 f}{\partial t^2}\Delta t^2 + \cdots \tag{TN14.7}$$

Now substitute the discrete form of (TN14.6), namely $\Delta S = a\,\Delta t + b\,\varepsilon\sqrt{\Delta t}$, and count orders. The squared term is

$$\Delta S^2 = a^2\Delta t^2 + 2ab\,\varepsilon\,\Delta t^{3/2} + b^2\varepsilon^2\,\Delta t \tag{TN14.8}$$

Of these three, the first two are $o(\Delta t)$ and vanish when summed, by (TN14.5). The third is $O(\Delta t)$ and, by Section 1.1, its sum converges to $\int b^2\,dt$ with no residual randomness. Hence

$$\Delta S^2 \;\longrightarrow\; b^2\,dt \tag{TN14.9}$$

The remaining second-order terms in (TN14.7) contain $\Delta S\,\Delta t = O(\Delta t^{3/2})$ and Δt^2, both of which vanish. Substituting (TN14.9) and (TN14.6) into (TN14.7) and passing to the limit gives **Itô's lemma**:

$$\boxed{df = \left(\frac{\partial f}{\partial S}a + \frac{\partial f}{\partial t} + \frac{1}{2}\frac{\partial^2 f}{\partial S^2}b^2\right)dt + \frac{\partial f}{\partial S}\,b\,dW} \tag{TN14.10}$$

Setting $a = \mu S$ and $b = \sigma S$ recovers equation (14.9) of the main text.

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The two-variable case. For $f(x, y, t)$ with $dx = a_1 dt + b_1 dW_1$, $dy = a_2 dt + b_2 dW_2$ and $dW_1 dW_2 = \rho\, dt$, the same expansion carries an additional cross term $f_{xy}\Delta x \Delta y$. By the argument of Section 1.1 applied to the product,

$$\sum_i \Delta x_i\, \Delta y_i \;\longrightarrow\; \int b_1 b_2 \rho\, dt \tag{TN14.11}$$

which produces the extra drift term $f_{xy} b_1 b_2 \rho$ in equation (14.20).

1.3 The Lognormal Solution

Chapter 14 asserts that geometric Brownian motion has the explicit solution used for simulation. Apply (TN14.10) to $G = \ln S$ with $a = \mu S$ and $b = \sigma S$:

$$\frac{\partial G}{\partial S} = \frac{1}{S}, \qquad \frac{\partial^2 G}{\partial S^2} = -\frac{1}{S^2}, \qquad \frac{\partial G}{\partial t} = 0$$

so

$$d \ln S = \left(\mu - \frac{\sigma^2}{2}\right) dt + \sigma\, dW \tag{TN14.12}$$

The coefficients are constant, so this integrates directly:

$$\ln S_T = \ln S_0 + \left(\mu - \frac{\sigma^2}{2}\right) T + \sigma\,(W_T - W_0) \tag{TN14.13}$$

and since $W_T - W_0 \sim N(0, T)$,

$$\ln S_T \sim N\Big(\ln S_0 + \Big(\mu - \tfrac{\sigma^2}{2}\Big)\, T,\;\; \sigma^2 T\Big) \tag{TN14.14}$$

Exponentiating (TN14.13) gives the exact simulation scheme of equation (14.6).

Moments. If $\ln X \sim N(m, s^2)$ then, completing the square in the defining integral,

$$\mathbb{E}\big[X^k\big] = \mathbb{E}\big[e^{k \ln X}\big] = e^{km + k^2 s^2/2} \tag{TN14.15}$$

Setting $k = 1$ gives $\mathbb{E}[X] = e^{m + s^2/2}$; setting $k = 2$ and subtracting the square of the mean gives the variance quoted in the main text. Applying (TN14.15) with $k = 1$ to (TN14.14),

$$\mathbb{E}[S_T] = \exp\Big(\ln S_0 + \Big(\mu - \tfrac{\sigma^2}{2}\Big)\, T + \tfrac{\sigma^2 T}{2}\Big) = S_0 e^{\mu T}$$

confirming that the Itô correction is exactly what makes the expected price grow at μ.

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1.4 Why the Measure Change Preserves Volatility

Chapter 14 states that moving to the risk-neutral measure changes a diffusion's drift but leaves its volatility untouched. The reason is visible from Section 1.1 without any formal apparatus.

A change of probability measure re-weights how likely each path is. It does not alter the paths themselves, and volatility is a property of a path rather than of its probability: by (TN14.4), the quadratic variation of S along any given path is $\int \sigma^2 S^2\, dt$, computed from the path alone. Since two equivalent measures agree on which paths are possible—they differ only in the weights—they must agree on quadratic variation, and therefore on σ.

The drift enjoys no such protection. It is defined by an *expectation*, and expectations are exactly what re-weighting changes.

The formal statement. Girsanov's theorem makes this precise. If W is a Wiener process under P and λ is a suitably regular process, then

$$W_t^Q = W_t + \int_0^t \lambda_u\, du \tag{TN14.16}$$

is a Wiener process under a measure Q equivalent to P, with the change of measure given by the density

$$\frac{dQ}{dP} = \exp\!\left(-\int_0^T \lambda_u\, dW_u - \frac{1}{2}\int_0^T \lambda_u^2\, du \right) \tag{TN14.17}$$

Substituting (TN14.16) into $dS = \mu S\, dt + \sigma S\, dW$ gives

$$dS = (\mu - \lambda\sigma) S\, dt + \sigma S\, dW^Q$$

so choosing $\lambda = (\mu - r)/\sigma$—the **market price of risk**—produces the risk-neutral dynamics $dS = rS\, dt + \sigma S\, dW^Q$. The volatility is unchanged, as claimed, and only the drift has moved.

The quantity λ deserves a name because it recurs: it is the excess return per unit of volatility, and Chapter 30 shows that changing numeraire amounts to changing λ, which is why convexity, timing, and quanto adjustments all take the form of a drift correction and never a volatility correction.

Further Reading

Neftci's *An Introduction to the Mathematics of Financial Derivatives* works through the arguments above at greater length and with more attention to what is being glossed over. Baxter and Rennie's *Financial Calculus* gives the shortest honest account of Girsanov's theorem and the martingale representation theorem. Readers wanting the measure-theoretic treatment should go to Karatzas and Shreve, or to Shreve's second volume, which is written for exactly this audience.

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